

Square roots and cube roots

A Cambridge insert: undoing a square and a cube, and knowing the small ones by heart.

LESSON 40

1 × 40 MIN

MATEMATIKA 6, QO'SHIMCHA MAVZU

S7 1.6

LESSON CLOCK

10'

12'

12'

6'

Squares and cubes	10 min
The roots	12 min
Estimating	12 min
In area and volume	6 min

LEARNING OBJECTIVES

- Find squares and cubes of small whole numbers.
- Find square roots and cube roots exactly where they are whole.
- Estimate a square root between two whole numbers.
- Use roots in area and volume problems.

TEXTBOOK

National	pp. 106–108
Cambridge	Stage 7, pp. 27–31
Workbook	Exercise 1.6

01

Explanation

Squares and cubes

<i>N</i>	1	2	3	4	5	6	7	8	9	10
<i>n</i> ²	1	4	9	16	25	36	49	64	81	100
<i>n</i> ³	1	8	27	64	125	216	343	512	729	1000

LEARN THESE TWO ROWS

They are needed for the rest of school mathematics — in area, in volume, in Pythagoras and in every quadratic equation from Grade 8 onwards. Ten minutes now saves hours later.

The roots

$\sqrt{25} = 5$ because $5^2 = 25$; $\sqrt[3]{27} = 3$ because $3^3 = 27$

ROOT	VALUE	CHECK
$\sqrt{49}$	7	$7^2 = 49$
$\sqrt{81}$	9	$9^2 = 81$
$\sqrt{100}$	10	$10^2 = 100$
$\sqrt[3]{8}$	2	$2^3 = 8$
$\sqrt[3]{125}$	5	$5^3 = 125$
$\sqrt[3]{1000}$	10	$10^3 = 1000$

A ROOT UNDOES A POWER; IT DOES NOT HALVE IT

$\sqrt{36}$ is 6, not 18. The question a root asks is “what number, multiplied by itself, gives this?” — never “what is half of it?”

Estimating

Most numbers are not perfect squares. Trap the root between the two nearest ones.

ROOT	BETWEEN	NEARER TO
$\sqrt{50}$	7 and 8	7
$\sqrt{30}$	5 and 6	5
$\sqrt{90}$	9 and 10	9
$\sqrt[3]{30}$	3 and 4	3

$\sqrt{50}$ IS A LITTLE OVER 7

Because 49 is just below 50 and 64 is far above it. Comparing the gaps tells you which whole number the root is nearer, without any calculation.

In area and volume

PROBLEM	WORKING	ANSWER
a square of area 64 cm^2 : its side	$\sqrt{64}$	8 cm
a square of area 121 cm^2 : its perimeter	$4 \cdot \sqrt{121}$	44 cm
a cube of volume 125 cm^3 : its edge	$\sqrt[3]{125}$	5 cm
a cube of edge 4 cm : its volume	4^3	64 cm^3
a cube of volume 64 cm^3 : its surface	$6 \cdot 4^2$	96 cm^2

SQUARES BELONG TO AREA, CUBES TO VOLUME

Two directions give cm^2 and a square; three give cm^3 and a cube. The units say which root is needed before any number is written.

02 Worked examples

EXAMPLE 1 Find $\sqrt{81}$ and $\sqrt[3]{64}$.

① $9^2 = 81$, so $\sqrt{81} = 9$.

② $4^3 = 64$, so $\sqrt[3]{64} = 4$.

Note that $\sqrt{64} = 8$ — different roots, different answers.

ANSWER

9 and 4

EXAMPLE 2 Between which two whole numbers does $\sqrt{50}$ lie?

- ① $7^2 = 49$ and $8^2 = 64$.
- ② $49 < 50 < 64$
- ③ So $7 < \sqrt{50} < 8$, and much nearer 7.

ANSWER

Between 7 and 8

EXAMPLE 3 A cube has volume 125 cm^3 . Find its edge and its surface area.

- ① $\sqrt[3]{125} = 5$, so the edge is 5 cm .
- ② Each face is $5^2 = 25 \text{ cm}^2$.
- ③ Six faces: 150 cm^2 .

ANSWER 5 cm and 150 cm^2 **03** Interactive model

Build a cube from 27 unit blocks and ask for its edge; the cube root is picked up in the hand rather than from a table.

A number, its square and its square root

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Square	Kvadrat	Квадрат
Cube	Kub	Куб
Square root	Kvadrat ildiz	Квадратный корень
Cube root	Kub ildiz	Кубический корень
Perfect square	To'la kvadrat	Полный квадрат
Perfect cube	To'la kub	Полный куб
To estimate	Baholash	Оценить
Inverse operation	Teskari amal	Обратное действие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $\sqrt{49}$ equals: $\sqrt{100}$ equals: $\sqrt[3]{27}$ equals:

$\sqrt[3]{64}$ equals:

4

8

16

32

$\sqrt{50}$ lies between:

6 and 7

7 and 8

8 and 9

24 and 25

A square of area 64 cm^2 has side:

8 cm

16 cm

32 cm

4 cm

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 6^2

ANSWER 36

2. 9^2

ANSWER 81

3. 4^3

ANSWER 64

4. 5^3

ANSWER 125

5. $\sqrt{49}$

ANSWER 7

6. $\sqrt{100}$

ANSWER 10

7. $\sqrt[3]{27}$

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. $\sqrt[3]{64}$

ANSWER 4

2. $\sqrt{64}$

ANSWER 8

3. $\sqrt[3]{1000}$

ANSWER 10

4. $\sqrt{50}$ lies between

ANSWER 7 and 8

5. $\sqrt[3]{30}$ lies between

ANSWER 3 and 4

6. A square of area 121 cm^2 : its perimeter

ANSWER 44 cm

7. A cube of volume 125 cm^3 : its edge

ANSWER 5 cm

Hard · stretch and reason

7 PROBLEMS

1. A cube of volume 64 cm^3 : its surface area

ANSWER 96 cm^2

2. A cube of volume 125 cm^3 : its surface area

ANSWER 150 cm^2

3. $\sqrt{144} + \sqrt[3]{8}$

ANSWER 14

4. $\sqrt{(36 + 64)}$

ANSWER 10

5. Is $\sqrt{36} + \sqrt{64}$ the same as $\sqrt{(36 + 64)}$?

ANSWER No — 14 against 10

6. A square of area 90 cm^2 : its side, to the nearest cm

ANSWER 9 cm (a little over 9.4)

7. Which is larger, $\sqrt{81}$ or $\sqrt[3]{512}$?

ANSWER $\sqrt{81} = 9$ beats $\sqrt[3]{512} = 8$

07 Homework

Homework — 5 tasks

Learn the squares to 12 and the cubes to 10 by heart.

- Write the squares of 1 to 12 .
- Write the cubes of 1 to 10 .
- Find $\sqrt{169}$, $\sqrt{225}$ and $\sqrt[3]{216}$.
- Between which whole numbers does $\sqrt{200}$ lie?
- A cube has volume 343 cm^3 . Find its edge and its surface area.